During the 7th to 10th weeks of gestation, while the diaphragm is completing its formation, defects can occur that leave a passageway in the diaphragm wall. This defect allows the intestines to intrude into the thoracic cavity with profound consequences for the cardiopulmonary function of the infant. About 90% of the defects are a posterolateral hernia (Bochdalek hernia). The incidence of congenital diaphragmatic hernia (CDH) is about 1 in 2000 live births. There is a 30% chance that an infant with Bochdalek's CDH will also have congenital heart defects and a 10% chance that chromosomal abnormalities will be present.

The initial presentation of CDH is respiratory distress due to lung hypoplasia with tachycardia and tachypnea being present at birth. Compression of the vena cava by the intestines can also produce reduced preload and a corresponding decrease in cardiac output. Pulmonary hypertension with right-to-left shunting can contribute to hypoxemia. The abdomen may appear scaphoid due to its contents being extruded into the thorax.



Congenital Diaphragmatic Hernia: loops of intestine can be seen in the left thorax

On recognition of the condition, the patient's airway should be secured. If hemodynamic status and time permit, analgesics may be administered to reduce catecholamine release due to laryngoscopy. Mask ventilation should be avoided to prevent worsening of the condition by forcing air into the stomach. A nasogastric tube should be placed to decompress the stomach.

Critical care management of the patient varies greatly due to the wide range of concurrent disorders and variability in severity of the problem. Therapies such as deliberate alkalosis to increase pulmonary blood flow and reduce ventilation-perfusion mismatching is sometimes performed. Nitric oxide may be used to treat pulmonary hypertension. Extracorporeal membrane oxygenation, intratracheal pulmonary ventilation, and high-frequency oscillatory ventilation have all been used successfully until the patient was stable enough for surgical repair of the hernia.

Anesthetic management for the surgical repair of the hernia is often a high-dose opioid technique with a small amount of volatile anesthetic. Nitrous is not recommended for this surgery because of its restriction on FiO2 and enlargement of air-filled intestinal spaces.

Cote CJ, Lerman J, & Anderson BJ. A Practice of Anesthesia for Infants and Children. 5th Ed. Philadelphia: Elsevier Saunders; 2013: 286, 756-757.

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